Experiment No. 6

Title

Exploring Virtual Terminals, Display Managers, X Clients, and Window Managers in Linux

Aim

To study and understand how Linux manages multiple user sessions through virtual terminals and graphical environments, and to practice switching between text-based and graphical interfaces for system administration and security operations.

Theory

Virtual Terminals

Linux supports multiple virtual terminals (VTs) that allow users to access several independent terminal sessions simultaneously. These terminals run in text mode and are useful for system recovery and troubleshooting.

Common key combinations:

- Ctrl + Alt + F1 to Ctrl + Alt + F6 – Text-based virtual terminals
- Ctrl + Alt + F7 (or F1 on some systems) – Graphical interface

Display Manager

A display manager provides the graphical login interface and manages user sessions. It starts the graphical server and handles authentication.

Common display managers:

- GDM (GNOME Display Manager)
- LightDM
- SDDM

X Window System

- The X Window System (X11) is responsible for handling graphical display, keyboard, and mouse input. It enables graphical applications (X clients) to run on a Linux system.

X Clients

X clients are graphical applications that communicate with the X server to display windows. Examples include:

- Terminal emulator
- •Web browser
- Text editor

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Window Manager

A window manager controls the placement, appearance, and behavior of windows on the screen. Examples:

- GNOME Shell
- KDE Plasma
- Xfce
- Openbox

Commands Used

Commands	Description
tty	Display current terminal
who	Show logged-in users
w	Show user activity
staetx	Start X window system
systemctl status	Check display manager status
ps	View running process
echo $DISPLAY	Display X session variable

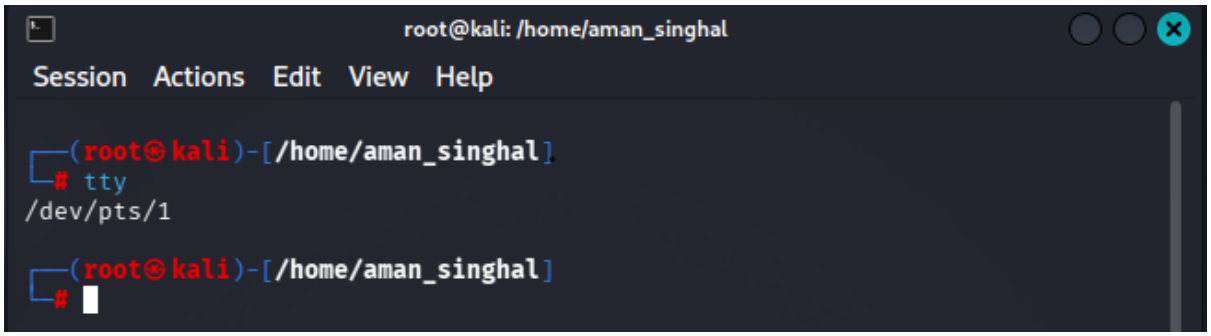
Procedure / Steps

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

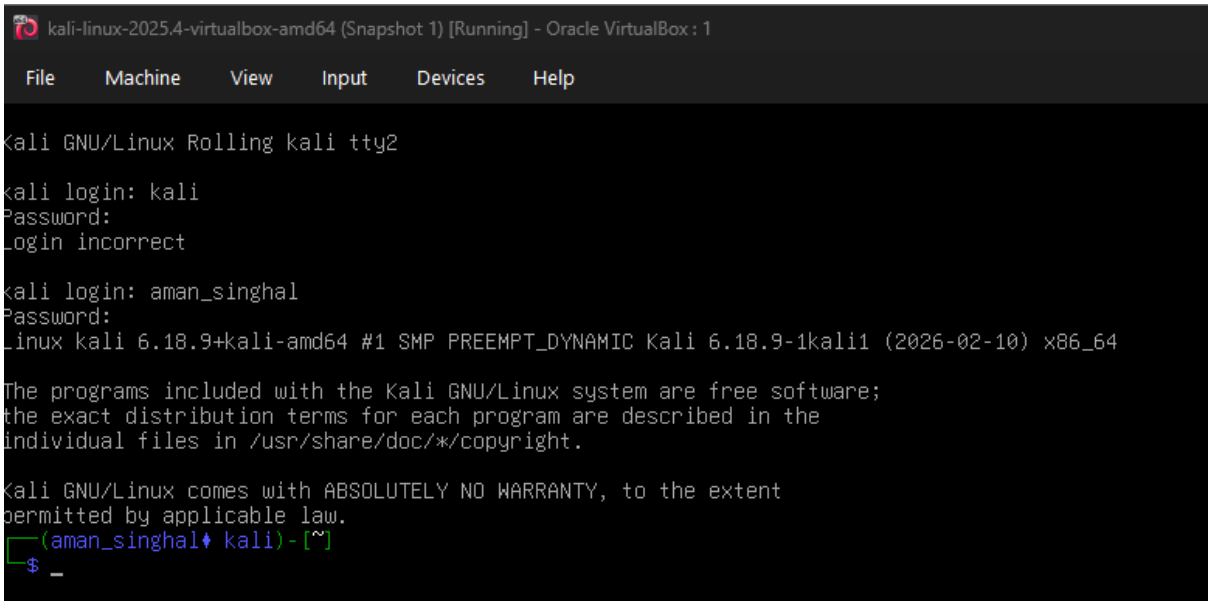
Step 2: Check the current terminal using:

- tty



Step 3: Switch to another virtual terminal using:

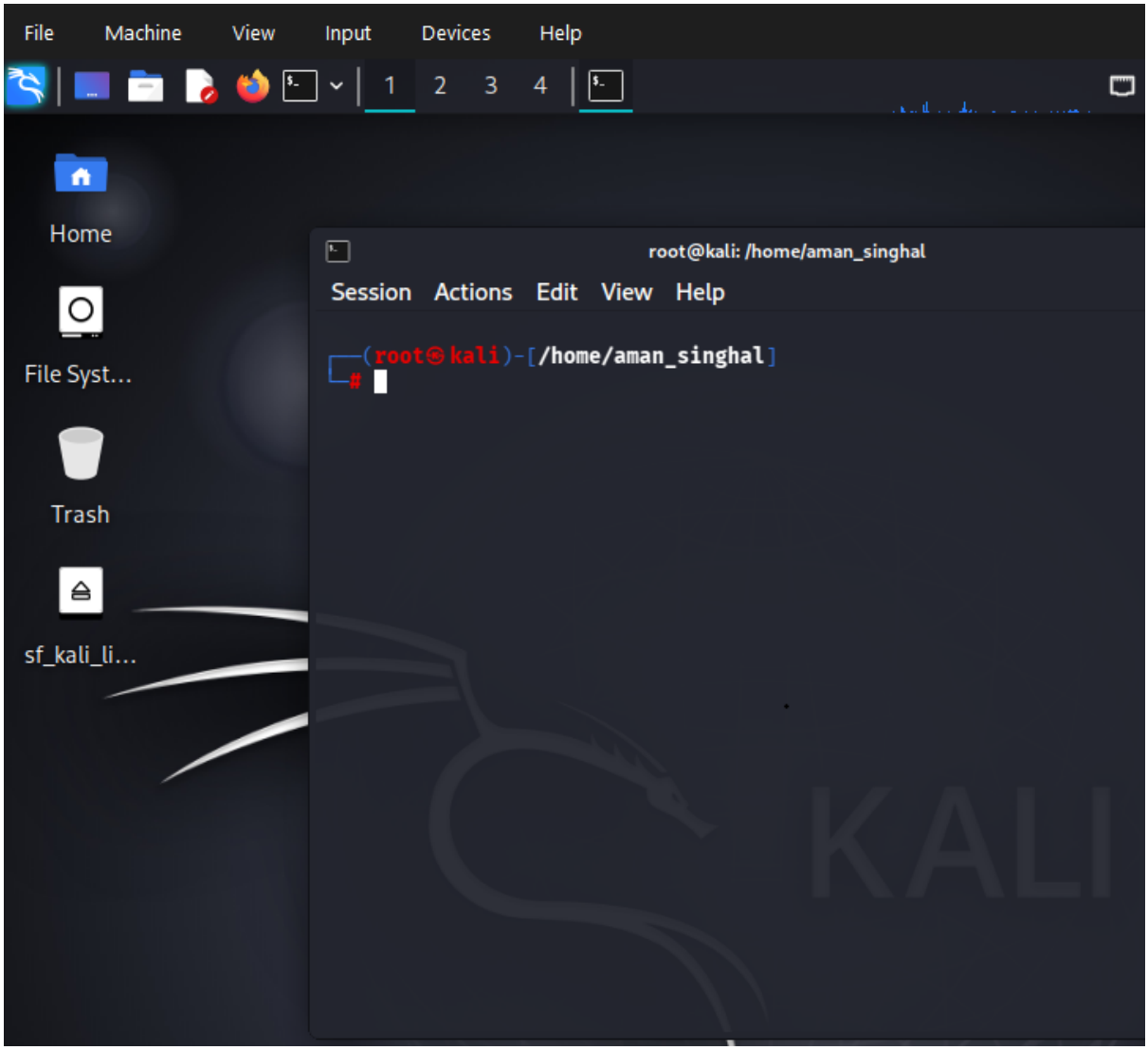
- Ctrl + Alt + F2



Step 4: Log in and verify terminal access.

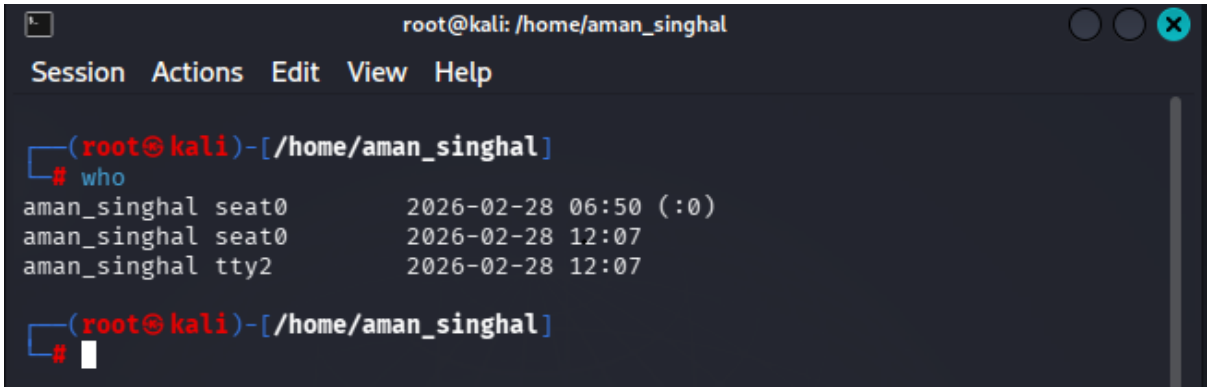
Switch back to graphical mode:

- `Ctrl + Alt + F1 or Ctrl + Alt + F7`



Step 5: Check logged-in users using:

- Who



Step 6: Check the display manager status using:

- systemctl status gdm

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help
● kali
    State: running
    Units: 295 loaded (incl. loaded aliases)
     Jobs: 0 queued
   Failed: 0 units
    Since: Sat 2026-02-28 06:43:47 EST; 5h 26min ago
  systemd: 259-1
  Tainted: unmerged-bin
   CGroup: /
           ├─init.scope
           │ └─1 /sbin/init splash
           ├─system.slice
           │ ├─ModemManager.service
           │ │ └─629 /usr/sbin/ModemManager
           │ ├─NetworkManager.service
           │ │ └─598 /usr/sbin/NetworkManager --no-daemon
           │ ├─accounts-daemon.service
           │ │ └─563 /usr/libexec/accounts-daemon
           │ ├─colord.service
           │ │ └─1257 /usr/libexec/colord
           │ ├─cron.service
           │ │ └─565 /usr/sbin/cron -f
           │ ├─dbus.service
           │ │ └─566 /usr/bin/dbus-daemon --system --address=systemd: --nofo>
           │ ├─haveged.service
           │ │ └─459 /usr/sbin/haveged --Foreground --verbose=1
           │ ├─lightdm.service
lines 1-27
```

Step 7: Verify X session variable Using:

- echo $DISPLAY

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help
┌──(root㉿kali)-[/home/aman_singhal]
└─# echo $DISPLAY
:0.0

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 8: List running X-related processes Using:

- `ps -e | grep X`

Outcomes

- ✓ Explain how Linux supports multi-user environments through virtual terminals.
- ✓ Differentiate between display servers, display managers, and window managers.
- ✓ Demonstrate switching between graphical and text-based sessions.
- ✓ Identify running X clients and window managers in a Linux system.
- ✓ Apply this knowledge in system administration tasks and cybersecurity operations such as session monitoring and privilege management.

Result

- ❖ Successful identification and use of multiple virtual terminals (Ctrl + Alt + F1–F6).
- ❖ Ability to switch between CLI and GUI environments.
- ❖ Clear understanding of the interaction between:

  - Display server
  - Display manager
  - Window manager
  - X clients

- ❖ Practical knowledge of managing Linux sessions for troubleshooting and security monitoring.